# Structural Transfer in Low-Resource Autoformalization

Runnel Zhang

METHODS • CONTROLLED STUDIES • VERIFIABLE EVIDENCE

A low-resource formal substrate exposes failures hidden by familiar syntax and library idioms. Diagnose the assembly of boundaries, dependent lifts, and nested mathematical witnesses.

Main deployment: gpt-5.4-2026-03-05 • temperature 0 • 1,200 completion tokens • no fine-tuning or theorem search

## 01 A structural microscope for local, verifier-grounded completion

### A. Freeze the information boundary

**Literate context**
local prose and recent formal prefix

→

**Target signature**
generate only the body after :=

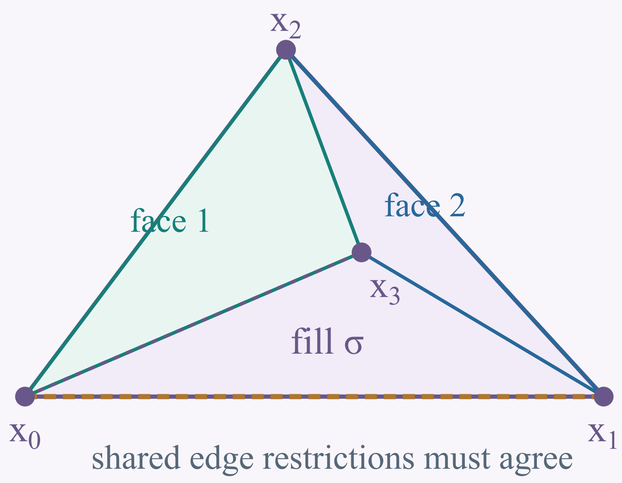
→

**Original source**
splice body back; run rzk typecheck

No future definitions in the prompt. Typechecking is the success criterion; the task isolates local structural construction.

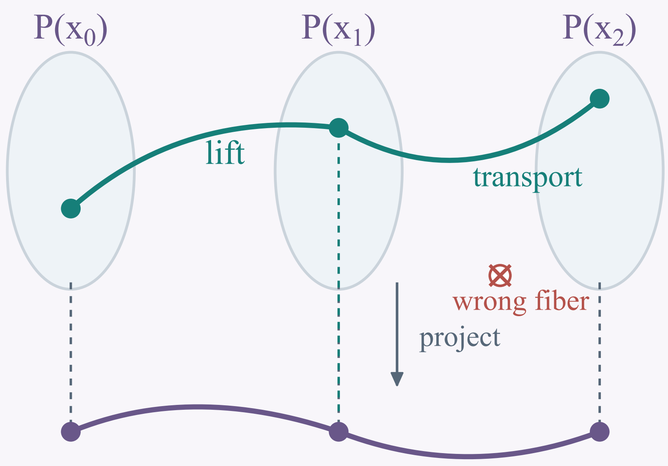
### B. Three interacting structural burdens

**BOUNDARY RECONSTRUCTION**



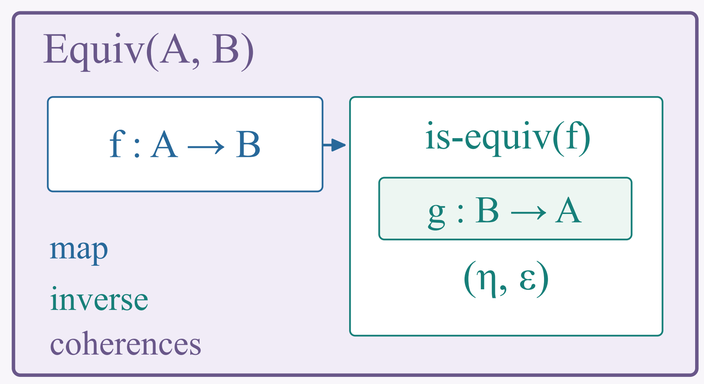

**Boundary depth**
endpoints → faces → horn layout

**DEPENDENT LIFT ASSEMBLY**




**Dependent depth**
hom → lift → fiber → fibration

**NESTED WITNESS PACKAGING**




**Witness nesting**
map → inverse → coherences

## 02 A layered benchmark

| Suite | n | Role |
|---|---|---|
| Easy Rzk controls | 8 | **pipeline check** |
| Shape-heavy dev. | 12 | **prompt design** |
| Held-out shapes | 28 | **diagnosis** |
| Same-family suite | 175 | **scale check** |
| Broad Rzk probe | 437 | **23 families** |
| Lean probe | 24 | **contrast** |

### Held-out object families

Extension scaffolds; Segal boundaries; covariant lifts; Yoneda / representable constructions; Rezk / isomorphism structures.

Boundary depth, witness nesting, and dependent depth are deterministic complexity annotations. They provide a structural lens on failure, beyond one pooled pass rate.

The 12-task development slice is reserved for prompt design. It does not enter the held-out claim.

## 03 Interventions on 28 tasks

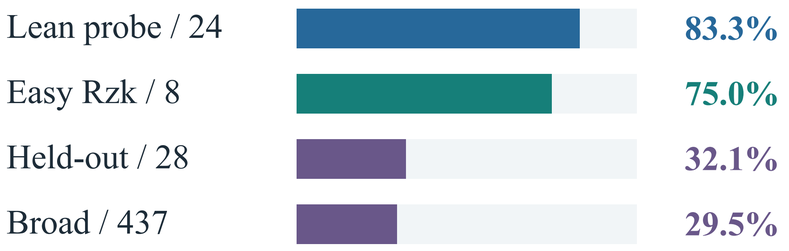
| Completion strategy | Solved | Rate |
|---|---|---|
| Direct | 9 | **32.1%** |
| Direct + repair | 14 | **50.0%** |
| Structure | 13 | **46.4%** |
| Structure-Lite | 14 | **50.0%** |
| Lite + repair | 15 | **53.6%** |
| Decomp-ICL | 14 | **50.0%** |
| Decomp-ICL + repair | 15 | **53.6%** |

Structure-Lite adds task-family routing, short construction guidance, and frozen targeted hints. A lightweight structural prior recovers a meaningful fraction of direct failures.

### Non-structural prompt controls

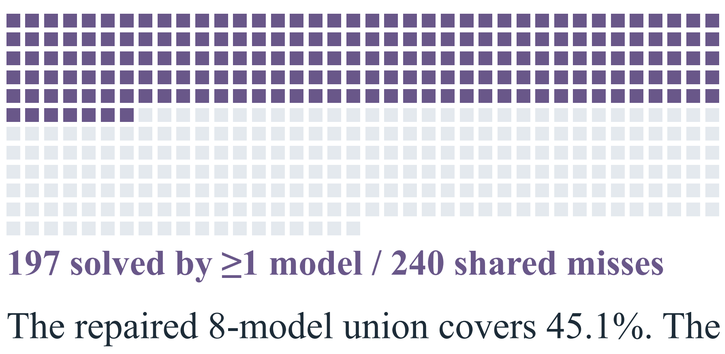
Skeleton-only prompts, length-matched neutral padding, pattern ICL, lexical ICL, same-family ICL, and decomposition ICL challenge simpler prompt-length or example-similarity explanations.

## 04 Transfer and shared failures




These probes are diagnostic contrasts, not difficulty-matched model rankings. Strong completion in a familiar or simple substrate does not ensure success on shape-explicit Rzk objects.

### 437-task broad-suite union



**197 solved by ≥1 model / 240 shared misses**

The repaired 8-model union covers 45.1%. The grid shows counts, not task ordering; 54.9% remain unresolved by every tested model.

## 05 Large-suite coverage

| Endpoint | 175 tasks | 437 tasks |
|---|---|---|
| Best one-shot | 29.7% | — |
| Best repaired | 34.3% | **36.4%** |
| 8-model union | 42.3% | **45.1%** |
| Shared misses | 101 | **240** |

Broad direct + repair: 154/437 = 35.2%.

## 06 Replay fixed failures

| Repair instruction | Solved | Rate |
|---|---|---|
| Original | 5/120 | **4.2%** |
| Compiler hint | 7/120 | **5.8%** |
| Structural | 11/120 | **9.2%** |

Freeze body, plan, examples, and compiler feedback; vary only repair-time instructions.

## 07 88 audited failures

